



Question 1 of 30

This is a photo of one of Mars' moons called Phobos.



This is a photo of Earth's moon.



How is Phobos like Earth's moon?

- ☐ It is reddish in colour.
- ☐ It has rings around it.
- ☐ It is as round as a ball.
- ☐ It has dents on its surface.



Question 2 of 30



Lucy tests this windmill by blowing at it through one straw at a time.

Click on the straw that makes the windmill spin in the opposite direction to the other three straws.



◀ Back

Flag 🚩

Next ▶

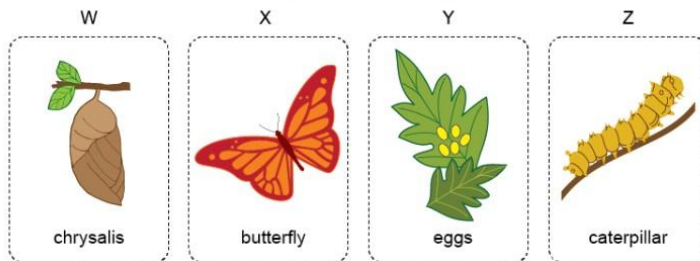


Question 3 of 30

There are four stages in the life cycle of a butterfly:

- Stage 1: A female butterfly lays her eggs on plant leaves.
- Stage 2: A caterpillar hatches from each egg.
- Stage 3: The caterpillar becomes a chrysalis.
- Stage 4: An adult butterfly comes out of the chrysalis.

These cards show the four stages, but not in the correct order.



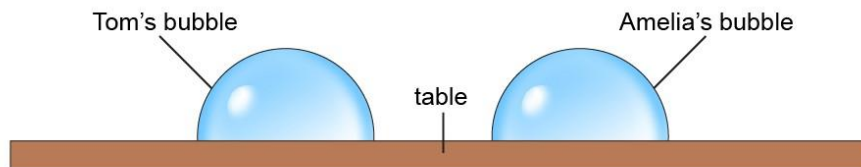
Which of these represents the correct order of stages for the life cycle of a butterfly?

- ☐ W → X → Y → Z
- ☐ X → Z → Y → W
- ☐ Y → Z → W → X
- ☐ Y → W → Z → X



Question 4 of 30

Tom and Amelia each blow a bubble onto this table.



They want to find out whose bubble lasts longer.

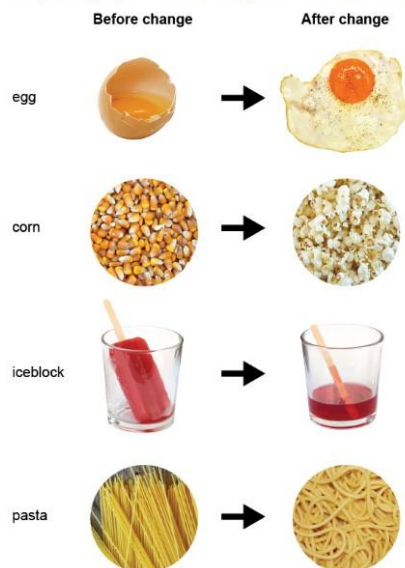
What should they do with their bubbles now?

- ☐ wait to see which one pops first
- ☐ blow them to get them to float
- ☐ wait to see if their colours change
- ☐ compare their size



Question 5 of 30

Some changes can be easily undone. Other changes cannot be undone.
The photographs show changes in some foods.



Which food can be easily changed back?

- ☐ egg
- ☐ corn
- ☐ iceblock
- ☐ pasta



Question 6 of 30

Which bird is likely to be the fastest runner?





Question 7 of 30

The table shows the weather forecast for the next few days.

Day	Weather	Minimum temperature (°C)	Maximum temperature (°C)
Monday		5	16
Tuesday		5	17
Wednesday		7	17
Thursday		6	19
Friday		3	18

KEY			
sunny	partly cloudy	overcast	rain

Which day is predicted to have the highest temperature?

☐

Monday

☐

Tuesday

☐

Wednesday

☐

Thursday

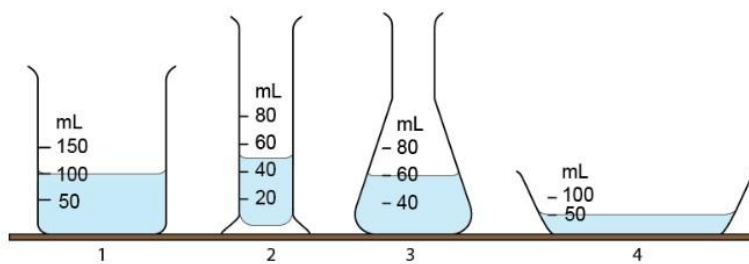
☐

Friday



Question 8 of 30

Mrs Black has four containers with water in them.



Select **two** containers that have the same amount of water.

☐

1

☐

2

☐

3

☐

4



Question 9 of 30

Dion cut an apple in half. He rubbed lemon juice onto the cut surface of one half only. He observed both halves an hour later.

The table shows his results.

	Before	After 1 hour
with lemon juice		
no lemon juice		

What could explain these results?

- ☐ Lemon juice stops the cut surface turning brown.
- ☐ Lemon juice stops the cut surface turning white.
- ☐ Lemon juice turns the cut surface brown.
- ☐ Lemon juice turns the cut surface white.



Question 10 of 30

Rob took this photo outdoors on a sunny day.



When he took the photo, four objects were lined up with the Sun.

Move the objects to line them up in the same order, starting with the Sun.

Sun

Rob

leaf

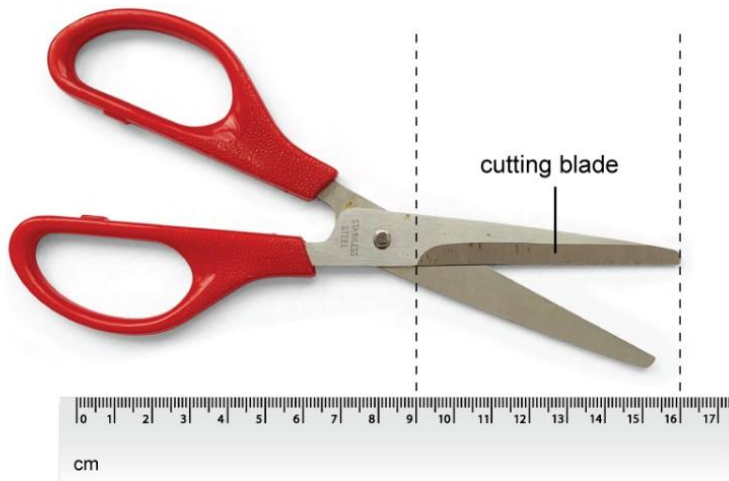
camera

bug



Question 11 of 30

These are Sean's scissors.



How many centimetres (cm) long is the cutting blade?



6



7



9



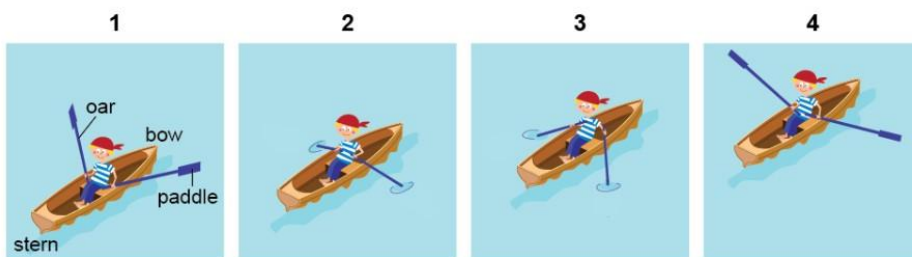
16



Question 12 of 30

Oars are used to row a boat. The bow is the front of the boat and the stern is the back.

These diagrams show how Tim uses oars to move his boat. He repeats this in the same order again and again.



Move the words to complete the sentences.

bow

stern

forwards

backwards

In the water, the paddles move towards the .

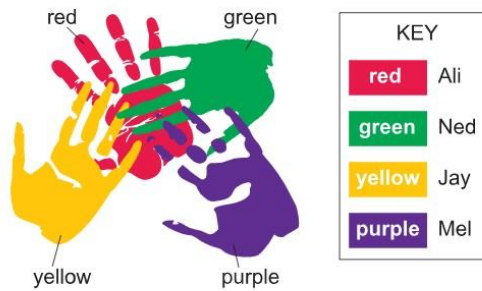
Above the water, the paddles move towards the .

The boat moves .



Question 13 of 30

Ali, Ned, Jay and Mel made an artwork using palm prints. They dipped one hand into the paint and then pressed their palms onto a piece of paper one after another.



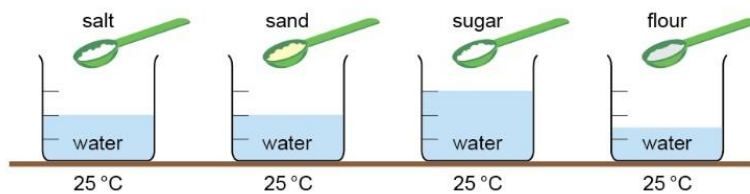
In which order did they press their hands on the paper? Select the correct row in the table.

1st	2nd	3rd	4th
Ali	Ned	Jay	Mel
Mel	Ned	Jay	Ali
Ali	Jay	Ned	Mel
Jay	Ali	Mel	Ned



Question 14 of 30

A group of students wanted to investigate the effect of mixing different solids with water. They set up their experiment as shown.



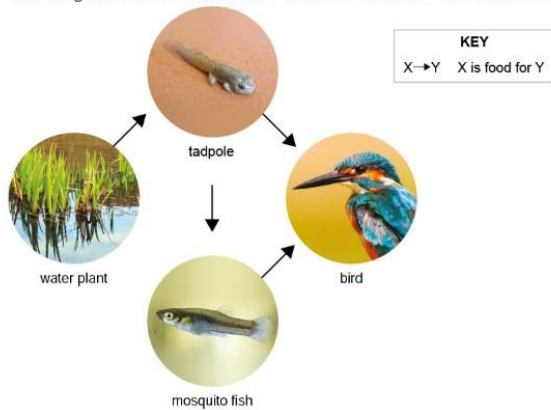
Which statement is correct?

- ☐ It was a fair test because the water temperature was the same in each beaker.
- ☐ It was not a fair test because the amount of water was not the same in each beaker.
- ☐ It was a fair test because the same amount of each solid was added in each beaker.
- ☐ It was not a fair test because all the solids were different colours.



Question 15 of 30

The diagram shows what some animals eat in the area where they live.



Which of the following is correct based on the diagram?

- ☐ Tadpoles eat mosquito fish.
- ☐ Mosquito fish eat water plants.
- ☐ Mosquito fish eat tadpoles and birds.
- ☐ Birds eat tadpoles and mosquito fish.

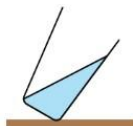


Question 16 of 30

The picture shows a glass with some water in it.



Which picture correctly shows the water level in the glass when it is tilted?



Question 17 of 30

Jay's mum is making some dough. She has some flour in a bowl. She tells Jay that she will need $1\frac{1}{2}$ cups of water.



Jay's mum then decides to double the amount of flour in the bowl to make more dough.
How many cups of water will she need altogether to make the dough?



$1\frac{1}{2}$



2



$2\frac{1}{2}$



3

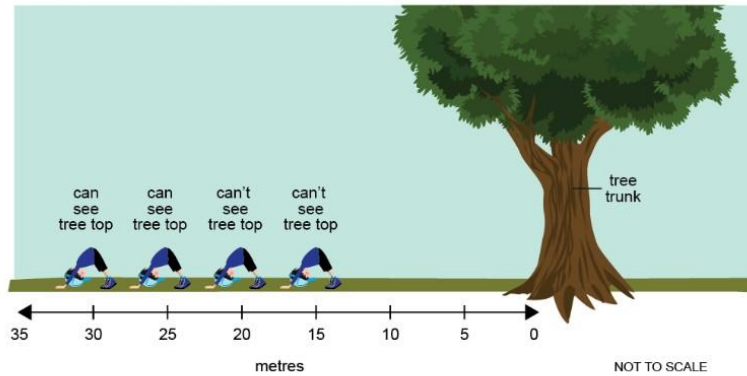


Question 18 of 30

Max stood close to a tree. He followed these instructions:

1. Bend over and look through your legs at a tree.
2. If you can't see the top, move away from the tree until you can.
3. You are now as far away from the tree as it is high.

This diagram shows Max's results.



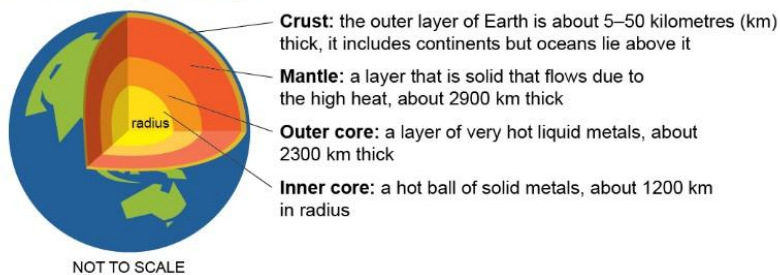
Based on this method, about how tall is the tree?

- ☐ 15 metres
- ☐ 20 metres
- ☐ 25 metres
- ☐ 30 metres



Question 19 of 30

Earth is made up of four layers.



Which statement is supported by the information?

- ☐ The mantle contains ocean floors.
- ☐ The inner core is a ball of liquid.
- ☐ The outer core is a solid layer surrounding the inner core.
- ☐ Earth mostly contains hot materials that flow.



Frogs hunt on land using their sight. An investigation showed that frogs will only try to eat dark objects smaller than themselves that move against the background.



Tick a box in each row to show which animals frogs will or will not try to eat.

	frogs will try to eat	frogs will not try to eat
living flies that move around them	☐	☐
dead moths on a large moving leaf	☐	☐
living mosquitoes that stay still	☐	☐
adult ducks that swim past them	☐	☐



Question 21 of 30



This table lists the time of sunrise in Sydney over seven days in May 2018.



Date	Sunrise (am)
May 23	6:45
May 24	6:46
May 25	6:46
May 26	6:47
May 27	6:48
May 28	6:48
May 29	6:49

Based on the information in the table, what is the most likely time of sunrise on May 31, 2018?

☐

6:49 am

☐

6:50 am

☐

6:51 am

☐

6:52 am

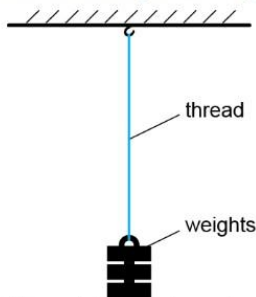


Question 22 of 30



Kerry tested the strength of four different types of thread.

She added weights to each piece of thread until the thread broke.



The table shows her results.

Type of thread	Breaking weight (N)
acrylic	2.5
cotton	0.25
nylon	0.50
silk	2.0

Move what Kerry changed and what she measured in her experiment into the boxes.

the breaking weight.

the length of the thread.

the type of thread.

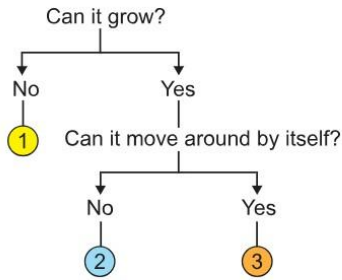
the height of the weights from ground.

Kerry changed

Kerry measured



Jessie had a plant, a cat and a toy car. Her dad drew this key to show her the differences between the three things.

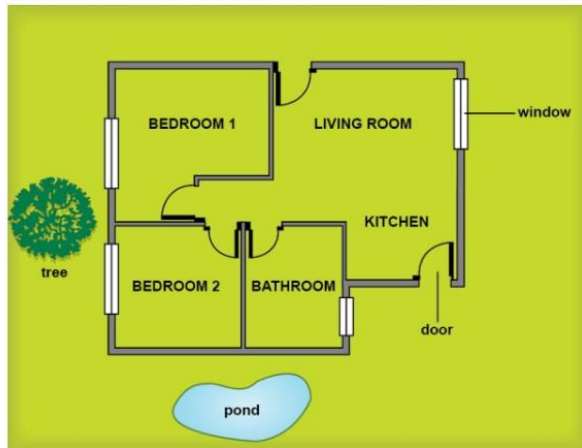


Select the row that correctly matches the numbers in the key with the three things Jessie had.

1	2	3
toy car	cat	plant
plant	cat	toy car
toy car	plant	cat
cat	toy car	plant



This is a plan of Siong's house as seen from above.



Siong stands inside the house. This is what she does next:

- She walks into a room. This takes her closer to the pond.
- She walks to a window.
- She sees a tree through the window.

Which room is she in?



bedroom 2



bedroom 1



bathroom



living room

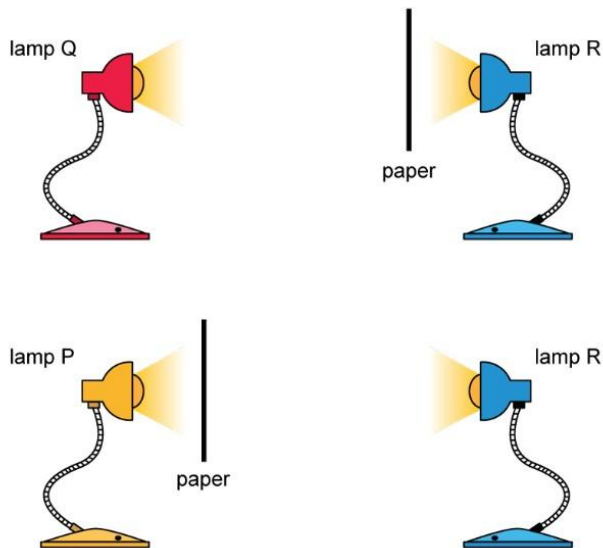


Question 25 of 30

The closer you are to a lamp, the brighter the lamp appears.

A spot of oil leaves a mark on a piece of paper. When the paper is lit with equal brightness on both sides, the oil spot seems to disappear.

Jai tested the brightness of two pairs of lamps using the paper with an oil spot. These diagrams show where the paper was between each pair of lamps when the oil spot disappeared.



Order the lamps from dimmest to brightest when they are observed from the same distance.

lamp P

lamp Q

lamp R

dimmest



brightest

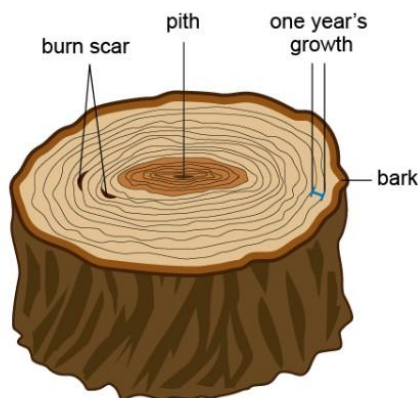


Question 26 of 30

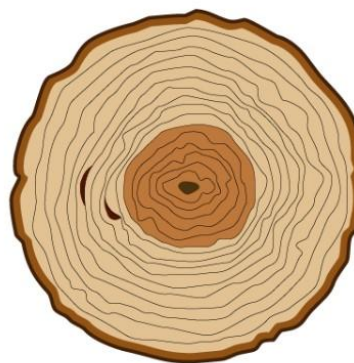
As a tree grows, it forms new wood just beneath the bark.

The diagram shows two views of a tree stump. Each ring represents one year's growth of wood. The pith is soft tissue found at the centre of the tree. A scar forms when a fire burns the tree's new growth.

side view of tree stump



top view of tree stump



How many years ago did the tree most recently get burnt? Type your number in the box below.

years

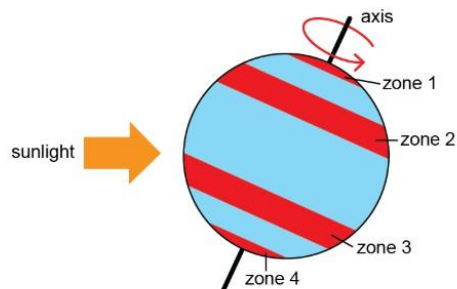


Question 27 of 30

Because Earth spins on its axis, different parts of Earth face towards the Sun as time passes. At any particular time, half of Earth's surface faces towards the Sun and has day time and the other half faces away from the Sun and has night time.

Different locations on Earth receive different numbers of daylight hours depending on the angle of the sunlight.

This diagram shows different zones on Earth at a particular time of year.



Draw lines to match the correct zone to each condition.

zone 1
zone 2
zone 3
zone 4

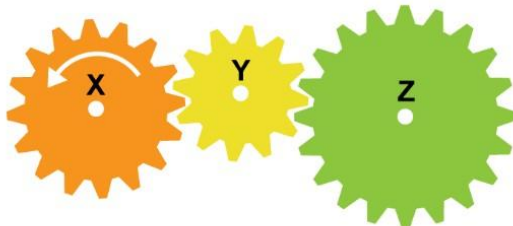
long days
short days
sun never sets
sun never rises



Question 28 of 30

Gears are wheels with teeth along their edge. The teeth on two gears can mesh so that one turning gear can make another gear turn without slipping.

This diagram shows three gears.



Gear X is turning anticlockwise.

How does gear Z move compared with gear X?

- ☐ It turns faster in a clockwise direction.
- ☐ It turns slower in a clockwise direction.
- ☐ It turns slower in an anticlockwise direction.
- ☐ It turns faster in an anticlockwise direction.

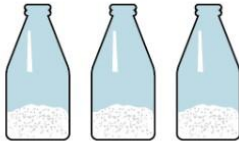


Question 29 of 30



Mr Belgrade had three bottles, each containing a different white solid. He knew that one bottle contained glucose (a type of sugar), one contained salt and one contained an unknown solid, X.

Mr Belgrade also had a liquid that turns red when added to a mixture of hot water and glucose.



Mr Belgrade knew that solid X turns blue when water is added to it.

He wrote down some steps to help him match each container with its contents. The steps are not in the correct order.

1. Take out a small sample of each of the solids.
2. Add some drops of the glucose-detecting liquid to each saucer to see which one contains glucose.
3. Add hot water to each saucer to find out which solid is X.
4. Place each sample into a separate saucer.

Which of these is the correct order of the steps to find out what each solid is?

- ☐ 1, 4, 2, 3
- ☐ 1, 4, 3, 2
- ☐ 4, 3, 1, 2
- ☐ 4, 2, 1, 3



Question 30 of 30



Hens living together form a pecking order. Each hen pecks only the hens below it in the pecking order.

Tony observed four hens called Sally, Preeti, Cilla and Helen.

His table shows how many times one hen (Hen 1) pecked another hen (Hen 2).

		Hen 2			
		Sally	Preeti	Cilla	Helen
Hen 1	Sally	-	0	0	0
	Preeti	5	-	0	0
	Cilla	22	35	-	14
	Helen	9	8	0	-

Move the names to order the hens into their pecking order.

Sally

Preeti

Cilla

Helen